Code No: 138FY **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, June - 2024
ELECTRONIC MEASURING INSTRUMENTS
(Common to CE, EEE, ME, CSE)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a) Explain the process of calibration. [2]
b) Define sensitivity. Calculate the sensitivity of a 200 μA meter movement which is to be used as a dc voltmeter. [3]
c) Define the function of wave analyser. [2]
d) Describe the fixed AF oscillator and variable AF oscillator. [3]
e) Describe the operation of an ohmmeter. [2]
f) Compare and contrast DC current meters and DC voltmeters. [3]
g) What is the basic operating principle of digital tape recording? [2]
h) Describe the functionalities of a curve tracer in electronic measuring instruments. [3]
i) What are the various characteristics of a transducer? [2]
j) Explain how an electrical transducer can be used to find the unknown pressure of a liquid. [3]

PART – B

(50 Marks)

2.a) A temperature sensor is known to have an accuracy of $\pm 0.5^0$C and a precision of $\pm 0.2^0$C. Determine the range within which the actual temperature lies with 95% confidence, considering both accuracy and precision.
b) Explain different types of errors that occur in measurements. [5+5]

OR

3.a) A D'Arsonval movement with a full scale deflection current of 50 μA and internal resistance of 500Ω is to be converted into a multirange voltmeter. Define the value of multiplier required for 0 -20v, 0-50v, 0-100v.
b) Explain the construction of multi-range voltmeter and Range extension DC voltmeter.[5+5]

4.a) An arbitrary waveform generator has a sample rate of 1 MS/s and 12-bit vertical resolution. Determine the maximum frequency that can be accurately represented in the generated waveform.
b) Discuss in detail about RF signal generator operation. [4+6]

OR

5.a) Design a square wave generator with an adjustable frequency ranging from 1 Hz to 1 kHz and an amplitude of 5 V peak-to-peak. Include the circuit diagram and explain the role of each component in generating the square waveform.

b) With a neat diagram, discuss the operation of a pulse generator. [6+4]

6.a) Discuss the design and functioning of AC voltmeters and current meters.

b) Explain with a diagram how frequency and phase can be measured using a Lissajous method. [6+4]

OR

7.a) Draw the block diagram of a general purpose oscilloscope (CRO) and explain function of each block in detail.

b) An AC voltmeter is designed to measure RMS voltages up to 250V at 50 Hz. Calculate the peak voltage and peak-to-peak voltage that this voltmeter can accurately measure.[6+4]

8.a) Explain the FM method of magnetic tape recording and explain its advantages and disadvantages.

b) Describe the closed-loop operation and the benefits of using servo transducers for accurate and reliable recordings. [5+5]

OR

9.a) Explain the working principle of an X-Y plotter used in electronic measuring instruments.

b) Describe the pen driving mechanisms employed in galvanometric recorders. [5+5]

10.a) With a neat sketch, explain the operation of LVDT.

b) Explain the piezoelectric effect and its utilization in piezoelectric transducers for measuring pressure, acceleration, and vibration. [5+5]

OR

11.a) Describe the operation of digital temperature sensing systems, and list it's advantages over analog methods.

b) Explain in detail about the resistive transducer with respective potentiometer. [5+5]

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